Engine Model(s): PT6A

74-10-01

IGNITION CURRENT REGULATOR - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire used shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, which is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC08-010	Cement, Adhesive
PWC11-023	Solvent, Cleaning
PWC11-025	Solvent, Cleaning

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

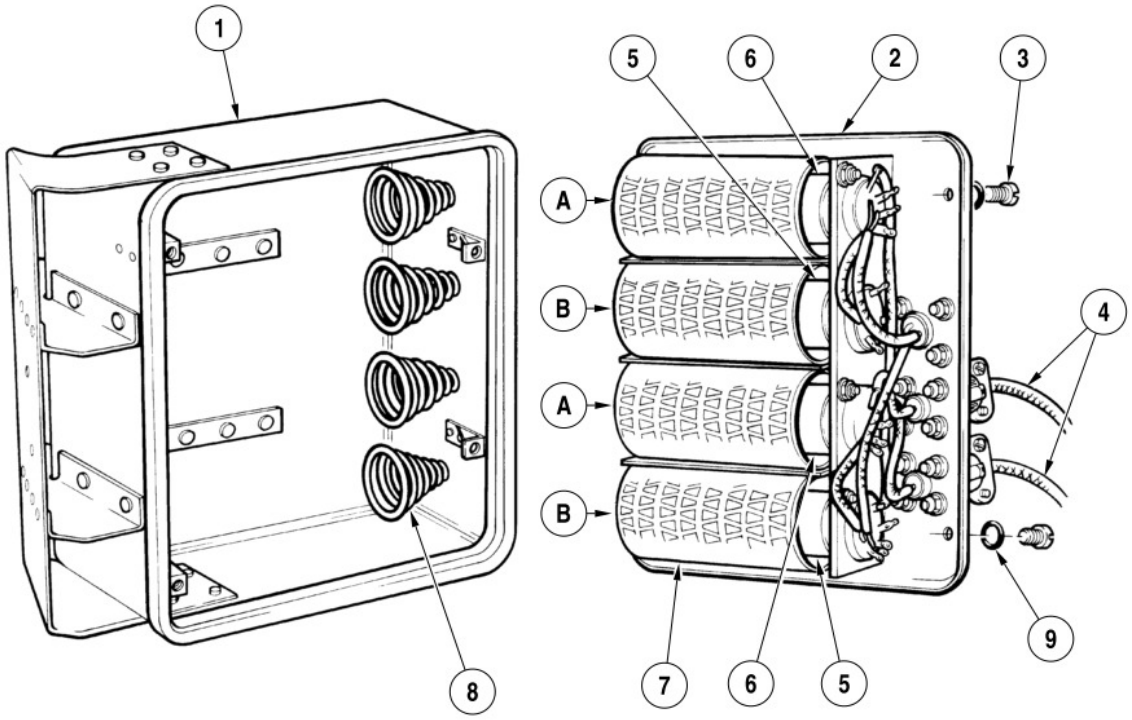
Not Applicable

5. Removal/Installation

A. Removal of Ignition Current Regulator (Ref. Fig. 201)

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Figure 201 Removal/Installation of Ignition Current Regulator



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1. Current Regulator Box
2. Current Regulator Cover
3. Screw
4. Ignition Cable
5. Ballast Tube (B)(P/N 3008040)
6. Ballast Tube (A)(P/N 3014055)
7. Tube Cushion
8. Compression Spring
9. Washer

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(1) Disconnect the airframe power supply cable from the receptacle on the current regulator (Ref. Airframe Manufacturer's Manual).

(2) Disconnect the ignition cables (4) from the receptacles on the current regulator.

(3) Remove the three bolts and washers and remove the current regulator from the accessory gearbox housing.

B. Installation of Ignition Current Regulator (Ref. Fig. 201)

(1) Locate the ignition current regulator on the three bosses on the accessory gearbox housing and secure with three washers and bolts. Tighten the bolts 36 to 40 lb.in., and secure with lockwire.

(2) Connect ignition cables (4) to respective receptacle on current regulator. Tighten connectors fingertight plus 45 degrees.

(3) Connect the airframe power supply cable to the receptacle on the current regulator (Ref. Airframe Manufacturer's Manual).

C. Replacement of Ballast Tubes (Ref. Fig. 201)

(1) Remove the ballast tubes:

(a) Disconnect the electrical and ignition cables (Ref. Subpara. A.).

(b) Remove the four screws (3) and carefully remove the cover (2) from the regulator box (1). Remove the preformed packings from the screws.

(c) Remove the ballast tubes (5) and (6) together with the cushions (7) from the receptacles in the cover.

(d) Remove the tubes from the respective cushions (7), exercising care not to damage the spring fingers.

(2) Install the ballast tubes:

(a) Install the cushions (7) on each tube (5 and 6).

(b) Locate the tubes with the cushions in the respective locations on the cover (2); press the pins in the base of the tubes firmly into the receptacles.

(c) Install the cover on the regulator box (1) by mating the tubes and compression springs (8), making sure the cushions (7) are not distorted.

(d) Install the preformed packings on the screws (3) and secure the cover with the screws. Tighten the screws 10 to 11.5 lb.in., and secure with lockwire.

(e) Connect the electrical supply and ignition cables (Ref. Subpara B.).

6. Cleaning/Painting

A. Cleaning of Box and Cover of the Ignition Current Regulator

(1) Remove the corrosion deposits on the exterior surface of the box and cover by light brushing.

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(2) Clean all components of the regulator box and cover, with quick-drying solvent (PWC11-023) or (PWC11-025). If necessary, dry using clean, compressed air.

(3) Make sure all receptacles are clear and clean.

7. Inspection/Check

A. Inspection of Ignition Current Regulator

(1) Inspect the cover and box of the regulator for general condition. A cracked or distorted mounting bracket on the box, or loose components on the box or cover, must be repaired at an overhaul facility.

(2) Inspect the seal on the box and the sealing gasket on the cover for general condition. A loose seal or gasket may be rebonded using adhesive cement (PWC08-010).

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